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***B.Tech. Degree IV Semester Special Supplementary Examination
September 2014***

**IT/CS 1405 DATA STRUCTURES AND ALGORITHMS
(2012 Scheme)**

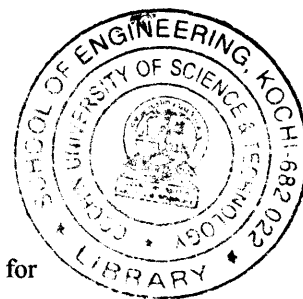
Time : 3 Hours

Maximum Marks : 100

PART A
(Answer *ALL* questions)

(8 × 5 = 40)

- I. (a) Differentiate between data structure, file structure and storage structure.
 (b) What are the significance of hash functions?
 (c) Explain the role of stack in postfix expression evaluation.
 (d) Define a Red-Black tree.
 (e) How are threaded binary trees represented in main memory?
 (f) Discuss any one method used to represent graphs.
 (g) Differentiate between B+ trees and B-trees.
 (h) What do you mean by 'minimum spanning tree'? Explain any one method for construction of minimum spanning tree.



PART B

(4 × 15 = 60)

- II. Give the algorithm for quick sort. Mention its time complexity. Explain it with an example. (15)
- OR**
- III. What are hash tables? Explain various hashing techniques. (15)
- IV. (a) What are linked lists? Discuss about various categories of linked list. (7)
 (b) Explain briefly how linked list can be used in polynomial manipulation. (8)
- OR**
- V. (a) Write the translation rules for converting an infix expression to prefix expression. (10)
 (b) What are sparse matrices? What are different representations? (5)
- VI. (a) What are the characteristics of AVL trees. (8)
 (b) What are threaded binary trees? Give its properties also. (7)
- OR**
- VII. Explain briefly how linked lists can be used for representation of binary trees. Also discuss about the various binary tree traversal techniques. (15)
- VIII. (a) Explain the Kruskal's algorithm. (7)
 (b) Write an algorithm for finding the shortest path in path ADT. (8)
- OR**
- IX. (a) Explain the various characteristics of a graph and how it can be represented using adjacency matrices and lists. (10)
 (b) Explain DFS and BFS in detail. (5)